

LearnWave — Master Blueprint

Comprehensive master blueprint covering features, pages, routing, Neon DB (Postgres), UI mapping, role-based routing (Student, Mentor, Admin), and Admin Portal. Sources: [filecite1turn0file0](#)
[filecite1turn0file1](#)

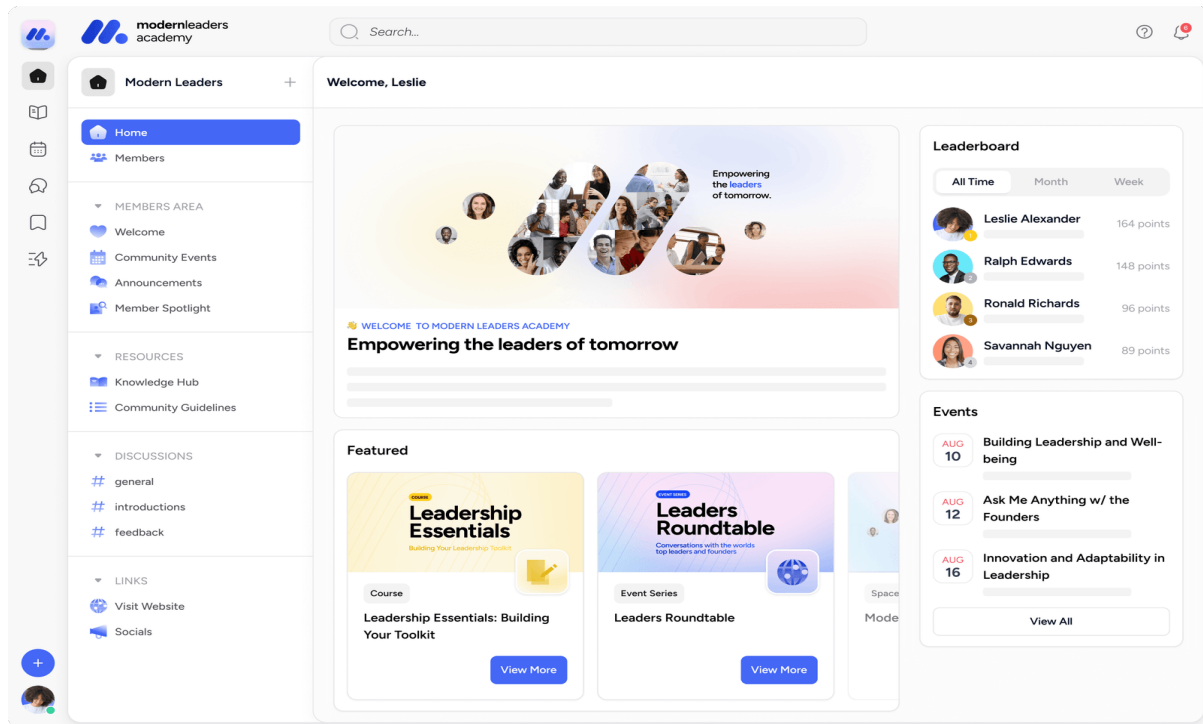


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1. Overview

This blueprint consolidates the uploaded specifications and the reference UI into an actionable implementation plan. It targets a polished, production-grade Next.js + Neon Postgres architecture with Prisma ORM, Tailwind UI, and a robust Admin portal. [filecite turn0file0](#) [filecite turn0file1](#)

2. Visual & UI Mapping

Color tokens, typography, motion system, and exact layout mapping to the provided reference image. Use Tailwind variables and shadcn/ui to match spacing, card radii, and subtle glassmorphism. Primary tokens: #4B7CFF, cream→pink→blue gradient.

3. Pages & Information Architecture

Exhaustive list of pages: public landing, auth flows, student (/s/*), mentor (/m/*), admin (/admin/*), resource viewer, forum, bookings, studio, analytics, notifications, settings. Each page includes mobile and desktop responsive variants.

4. Routing & RBAC

Use Next.js App Router with folder-based grouping and server middleware for RBAC checks. Protect /admin routes with server-side role verification; use session tokens for API calls.

5. Neon (Postgres) Schema — Highlights

Core models: User, Student, Mentor, Resource, ForumThread, Post, Booking, Notification, AuditLog. Use JSONB for flexible metadata; GIN indexes for search.

6. API Endpoints (Representative)

Auth: /api/auth/* | Users: /api/users/* | Admin: /api/admin/* | Resources: /api/resources/* | Forum: /api/forum/* | Bookings: /api/bookings/* | AI: /api/ai/*

7. Frontend Components & State

Component list: TopNav, SideNav, HeroCard, FeaturedCard, Leaderboard, EventsPanel, ResourceViewer, ForumThread, MentorStudio. State: Zustand for UI, TanStack Query for server data.

8. Admin Portal

Admin capabilities: mentor approvals, user CRUD, resource moderation, analytics, payments & payouts, audit logs, feature flags, impersonation, bulk actions and exports.

9. Security & Privacy

RBAC, signed S3 uploads, virus scanning, rate limiting, encryption for PII, audit logs, GDPR endpoints.

10. Performance & CI/CD

Lighthouse targets, Redis caching, CDN, GitHub Actions, tests with Vitest & Playwright, monitoring with Sentry & Prometheus.

11. Deployment & Infra

Neon Postgres DB, Vercel for frontend, Render or ECS for backend, S3 for storage, Stripe/Razorpay for payments. Automated backups and secrets management.

12. Accessibility & i18n

WCAG AA, ARIA roles, keyboard nav, RTL and multi-locale support with Next.js i18n.

13. Roadmap (8 weeks)

Week-by-week plan from auth & DB to AI features, analytics, QA and launch.

14. Appendix: Prisma snippet

```
generator client { provider = "prisma-client-js" } datasource db { provider = "postgresql" url =  
env("DATABASE_URL") } model User { id String @id @default(cuid()) email String @unique password  
String name String? role String @default("student") avatarUrl String? createdAt DateTime  
@default(now()) Student Student? Mentor Mentor? }
```

Document generated programmatically. Refer to original spec files for more (file citations above).